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Tools for Trusting Information in a World of Information Ov

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The internet has vastly increased the speed and spread of information, connecting the world in ways never before thought possible.

One of the main advantages of developing technology is that a wealth of information and data is freely available to anyone with an internet connection. One downside to this increase in speed and communication is the ease with which misinformation can be disguised and accepted as fact. Often, so much information is coming at us that it is difficult to know what information to believe and trust.

This article will explore some of the steps that we can take to better understand the information around us and test its validity. Some of these techniques and ideas may already be familiar to you from your own personal research. This article aims to increase information literacy and stimulate deeper thought about what information and news we accept as true. To that end, you are encouraged to explore, test and challenge these suggestions as well.

Truth, Facts, and What It Means for Something to Be True

For this discussion, we will talk about truth as simply being whether a statement or claim is accurate (truthful) or inaccurate (false). When we are doing research, whether for a class or reading the news, we generally want to determine if the facts or arguments being presented are reliable and accurate. There are a number of things we need to keep in mind when making that decision.

Not everything is either true or false

Some information can neither be true nor false. This includes things like feelings and opinions because they are not based on facts. Feelings and opinions are real but they are subjective. Statements such as “Dave is ugly,” “chocolate cake is better than carrot cake” or “I hate Martha” tell us about how the speaker feels but not about facts.

Other things can be either true or false. This includes things like research statistics or facts about the world and events that have taken place. For something to be true or false, there must be some way to check.

For example, you can check to find out if the statement “chocolate cake has more calcium than carrot cake” is correct. You cannot check to find out which is objectively better because that is a matter of opinion.

Skepticism, Ask Questions

If we want to start being more confident about the information we see, we (unfortunately) all need to be a little skeptical. If something stands out as being either too good or too bad to be true, do more research because it often is. When it comes to accepting new information, it is healthy to be skeptical, to demand answers and seek to understand and analyze the data.

Sources

One of the first and most important things we can do when we get information is to follow the adage and consider the source. Think about where you got the information. What were the source's possible motivations and how's their past reliability? Some sources seek to give us knowledge and facts about something to educate us. Other sources are motivated to convince us about something, either because they feel strongly about it or because they have something to gain by convincing us.

If someone's motivation means that they have something to gain by convincing you of something, they may not be trustworthy. If a speaker company puts out a report that says your ears will be healthier if you use their new technology, you need to find someone who isn't trying to sell you speakers and find out what they think about the claim.

Knowing what a source's motivation is can tell you a lot about how reliable the information is. The same goes for the reliability and expertise of the source. Sources with a history of being trustworthy or having particular expertise have a greater level of reliability.

Imagine you have two friends, one is always known to tell the truth, and the other is known to sometimes make up stories for excitement or attention. Which friend are you more likely to believe when they come and tell you about something important that is happening?

Now imagine both of your friends are known for being truthful; one of those friends has a degree in physics and the other has a degree in journalism. Imagine that you are learning to play billiards. Which friends' advice about how the ball will move around the table will help you win?

If possible, try to determine your source's motivation, expertise and reputation. Reserve your trust for those with a history of reliability or relevant special knowledge or training.

Primary and Lesser Information Sources

When we are doing research, there are two kinds of sources. Principal sources are generally considered reliable sources of information. News outlets like Reuters and the Associated Press employ teams of their own fact-checkers before issuing press releases. Similarly, research and studies that have been peer-reviewed can often offer reliable information. The Center for Disease Control, World Health Organization, and similar organizations are also generally known to be reliable sources. These are the sorts of sources that your teacher lets you use on your research papers.

On the other hand, teachers rarely accept sources such as anonymous blogs, Wikipedia, forums or discussion post entries, or YouTube videos. These are lesser sources because they do not come with the same level of reliability as Principal sources. If you are getting your information from a lesser source, it is even more essential to find the author's original Principal sources to check them for yourself.

Lesser sources are not always inaccurate or unreliable. Still, the ease of posting information online and a lack of significant information vetting on these sorts of sites means that misinformation and lies are much more likely to go unnoticed. Lesser sources should always prompt additional research from more reliable sources.

Analyzing and Checking Information

Anyone, anywhere in the world, can make a meme, a video, or a blog post, put made-up numbers or statistics on it, and post it online. As students and educated citizens, it is our job to be aware of these risks and to dig deeper into the information we see.

Checking Dates

Check to see the original date that the information was published. Sometimes old information is recirculated as new. In areas where information moves quickly, like information technology or medicine, a story that is several years old may no longer be accurate. Sometimes older opinion pieces don't take critical systemic changes into account because they predate those changes.

With so much information coming at us daily, the information's age can play a huge role in determining how valuable and reliable the information is today.

Language Clues

Reliable information should generally not seek to excite you, but rather it should prove things to you by presenting and supporting facts. Sensational language, statements like "unbelievable," "shocking," or "miraculous" should make you cautious. Always be suspicious if someone says things like, "no one else will tell you this," or makes specific claims about numbers, statistics, or research but doesn't tell you precisely where they got their information.

Keep in mind any news that you read that makes clear the writer's opinions should be fact-checked. Reliable opinion sources will tell you where they got their information, avoid exciting language, and make checking their sources relatively easy.

Ideally, any information you accept should be available from at least two Principal sources that you trust. If only one source is telling you something, be cautious. Journalists, real and self-proclaimed, at every level, make their careers by covering stories. If no one else covers the information, there is probably an excellent reason for that and it is very very rarely an oversight or a product of corruption. Very few information or news sources have secret inside information that is available only to them.

Fact-Checking Sources

The simplest way to fact-check sources is by clicking links if they are included with the information. If links go to Principal sources that you trust, the hunt is over. If links aren't included with the information, check the article's or post's end to see if they have been included there. If no links are included, a web search for the information may turn up Principal sources you can rely on or it may not.

If your search does not turn up any additional, reliable sources of the information, you may not be able to accept it as being true. On the other hand, if you turn up contradictory information from equal reliability sources, you may need to do additional research to determine which is accurate.

Google Research Tools

Google has several tools that can be extremely useful in fact-checking sources. First, Google Trends will let you explore your search term. It will tell you how common the search is, when and where most searches occur, and so on. This can help you to determine if something is new information or if it is just being recirculated.

Another valuable tool is the reverse image search (with more details on using the very simple tool here). By searching for photos from articles or even memes, you can identify when the picture first showed up online and see other places it has been used. Sometimes this can help you determine if a photograph has been changed or altered, or if it is being misrepresented.

Research and Reading Studies

Sometimes, when we're researching for a class or following up on the sources for other information, we will need to look at posted research studies. One of the most important things you can do when you read a study is to look at who funded it to determine the motivation of the creators. This is usually listed near the top of a research study under the names of the researcher(s). It tells you who paid for the research.

Whether it was a university or the candle wax manufacturers of America could make a massive difference in how much you should trust the information. The university should be seeking to gain knowledge, while a private company or organization may have a vested interest in convincing you of or selling you something.

Size of a Study and Participants

The other thing that you should always look at is the sample size and make-up of the participants. A study of only several hundred people is very unlikely to be a good representation of society.

Also, studies that consist of groups that are not diverse enough (including everything from gender and race, to economic class or physical location) will tell us about the included groups rather than society as a whole. Studies with lots of participants from a broad range of places and backgrounds will give you the most reliable information about the general population.

For example, if 100 women in rural Ireland all experience headaches resulting from stress, that is interesting but it tells us very little about the average woman in the world. Be cautious with research that claims limited groups are representative of broad social norms.

Explore the Methods and Outcomes

Finally, always look at what the authors of a study have to say about what they found. If you are using the study to fact-check a source, ensure that the source's explanation of the study's findings matches what the study itself says.

Basic Conclusion

Vetting information isn't always easy, but the feeling of trust knowing the information you are using and sharing is reliable makes the effort worth it. Considering these elements of your information will increase your ability to trust new information:

Date of original publication

Who is the source?

Motivation, reputation, and expertise of the source

Language Clues

Fact-Checking

More Advanced Tricks: Logical Fallacies to Look Out For

One of the most frustrating parts of deciding if information is reliable is that accurate information and data can be used in misleading ways. Here, we will briefly explore some of the most common ways that even accurate information can be used to present a false or inaccurate picture of the facts. We will explore just a few of the most common ways that information can be used misleadingly. These are things to be on the lookout for when gaining new information.

Emotional Appeals and Confirmation Bias

Sometimes we see something, online or offline, and it starts a fire in our belly. You know what I mean; it's that thing you see or read that makes you so sad, angry or disgusted, so you call your mom, best friend or partner to tell them about the unbelievable thing you just saw. Sometimes this is a fair and reasonable response to tragic or terrible information.

Unfortunately, this is also an easy way to trick us into failing to think or look more deeply at the information. If something makes you feel emotional when you read it, you should be cautious. Check on the information and find Principal sources that confirm the information before you accept it.

A poster saying "10 billion kids a year are kidnapped" is memorable and shocking, but it is also wholly made-up. Just as the statement "spending 6 months a year on vacation can extend your life by up to 100 years." Maybe you want it to be true (it feels good to think about) but I just made it up on the spot. Anyone can do that.

This is also true if the information agrees with what you already believe or think that you know. Confirmation bias is another trigger that stops us from thinking more in-depth about something. If we believe that the next neighborhood over is much more dangerous than our own, then we see someone post information saying that that neighborhood is the most dangerous part of the city, we might accept it right away.

It seems easy to agree with what we already know; it feels right. However, that right feeling means that we might fail to check on the information, and perhaps that neighborhood is not even in the top 50% of our city's dangerous areas. Our feelings convinced us to believe a lie.

If something seems or feels right, it is still crucial to check it. Find sources, do research and always remember that something seeming true is not the same thing as it being true.

Straw Man Arguments and False Dichotomies

One of the most common ways people attempt to prove a point is with a straw man argument. This means that they present an oversimplified, easy to attack, straw man version of their opponent's argument, which they can then rip apart.

For example, imagine a university study that says, "oranges have more vitamin C than lemons." Now, imagine a lemon farmer got mad about this and said: "that university is trying to get people to stop eating lemons to boost oranges' sales. They think people who eat lemons are less healthy than orange eaters because they get less vitamin C."

Strawman arguments take place often, but it's important to remember that they misrepresent the opponent's views and should not be trusted as sources of information. Understanding the position of someone you disagree with requires to listen to what they actually say and it takes more effort than simple misrepresentation.

Another trick to be careful with is the false dichotomy. Sometimes someone will try to make you feel like situations are black and white to get you on their side.

If your friend wants you to go out and says, “you have to come with me; otherwise, you’ll just be bored at home,” they fail to take into account the million other things you could do instead. Maybe you won’t be bored but busy working or having fun at home. Perhaps you won’t choose to stay home and instead will go to the coffee shop down the street. Similar arguments are often made seeking to gain support, “choose A, or you will get B.”

Right now, it’s pervasive to see people doing this politically, “if you don’t like candidate A you must love candidate B.” In fact, there is a broad range of ways to feel about all candidates, and perhaps you don’t like any of the options. These dichotomies are dangerous because they oversimplify ideas and sometimes pressure us to agree with things we otherwise would not to avoid being on the wrong side.

Imagine a poster saying; “Join our group. We’re dedicated to stopping climate change by blowing up everyone’s cars. Only people who hate the planet wouldn’t want to do this to save the environment.” This is a very extreme example, but you can see how you could be put in a position where you may feel pressure to accept or defend an idea that you usually wouldn’t, even one that could do more harm than good, to prove you do not hate the planet.

Correlation Versus Causation and the Burden of Proof

One of the most important things to remember when reading or researching new information is that correlation is not causation. This fact is easily and often forgotten. Consider the fact that everyone who breathes air and drinks water will eventually die. Does this mean that they die because they breathed and drank water? Of course not.

If a study showed that all students at the university who got an A last semester had eaten chocolate cake at their birthday party, would you assume that chocolate cake on a particular day was why they got the grade?

Ask yourself the same question when assessing other information. Often people with green eyes have red hair, but green eyes do not cause red hair. Instead, both are affected by the level of melanin a person produces. Ask yourself if the cause and effect being presented could instead be a correlation, which, while both true, could share another cause or even be unrelated?

It’s also important to remember that the burden of proof always belongs to the person trying to convince someone of something. Sometimes, people will present an argument and argue that you must prove that they are wrong. An example: Aliens live among us on earth in disguise. You can’t prove they don’t, so it’s obviously true.

The burden of proof always falls on the person who presents the theory or idea. If you want to tell your friends that the Yeti is real, you need to provide all of the evidence and information to prove that you are correct. It is never reasonably up to your friends to go out in the woods and prove that the Yeti isn’t real to win the argument.

Conclusions

Information is powerful. It surrounds us constantly as citizens (and as students) in a way people a few hundred years ago could not even imagine. In this day and age, as educated people, one of our jobs is to learn how to determine if that information is reliable.

Sharing misinformation with our friends and family does them a disservice. After all the work we have done in school, we want our reputations to be that of educated and reliable individuals. By taking the time to consider the information we see and hear, we can be more confident in what we think and believe; we can be more confident in the information we share with others.